

Bayes' Theorem: A Foundation of Probabilistic Reasoning

Abstract

Bayes' theorem describes how to update beliefs in light of new evidence. First formulated by Thomas Bayes in the 18th century and later developed by Pierre-Simon Laplace, it has become a cornerstone of statistics, machine learning, and scientific inference. This document presents the theorem, its derivation, key interpretations, and several illustrative applications including medical testing and spam filtering.

1. Introduction

Probability theory provides a rigorous language for reasoning under uncertainty. Among its most powerful tools is Bayes' theorem, which relates the conditional probability of a hypothesis given observed data to the prior probability of the hypothesis and the likelihood of the data. In an era of data-driven decision-making, Bayesian reasoning underlies applications ranging from clinical trials to autonomous vehicles.

The central question Bayes' theorem answers is: *given that we have observed some evidence, how should we rationally revise our belief about a hypothesis?*

2. Mathematical Formulation

2.1 Conditional Probability

The conditional probability of event A given event B is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

By symmetry, $P(B | A) = \frac{P(A \cap B)}{P(A)}$, so:

$$P(A \cap B) = P(A | B) P(B) = P(B | A) P(A)$$

2.2 Bayes' Theorem

Rearranging the identity above yields **Bayes' theorem**:

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

In the context of hypothesis testing, we write H for a hypothesis and D for observed data:

$$P(H \mid D) = \frac{P(D \mid H) P(H)}{P(D)}$$

The terms have well-known names:

- $P(H)$ — the **prior**: our belief in H before seeing D
- $P(D \mid H)$ — the **likelihood**: how probable the data are if H is true
- $P(H \mid D)$ — the **posterior**: our updated belief after seeing D
- $P(D)$ — the **marginal likelihood** (evidence): a normalising constant

2.3 The Law of Total Probability

When H takes discrete values $H_1, H_2, \dots, H_n$ that partition the sample space, the denominator expands via the law of total probability:

$$P(D) = \sum_{i=1}^n P(D \mid H_i) P(H_i)$$

This gives the full form:

$$P(H_k \mid D) = \frac{P(D \mid H_k) P(H_k)}{\sum_{i=1}^n P(D \mid H_i) P(H_i)}$$

3. Applications

3.1 Medical Diagnostic Testing

Consider a disease affecting 1% of a population. A test has 99% sensitivity (true-positive rate) and 95% specificity (true-negative rate, so 5% false-positive rate). Given a positive test result, what is the probability the patient has the disease?

Let H = “patient has disease” and D = “test is positive”:

$$P(H) = 0.01, \quad P(D \mid H) = 0.99, \quad P(D \mid \neg H) = 0.05$$

$$P(D) = 0.99 \times 0.01 + 0.05 \times 0.99 = 0.0099 + 0.0495 = 0.0594$$

$$P(H \mid D) = \frac{0.99 \times 0.01}{0.0594} \approx 0.167$$

Despite the accurate test, only about 16.7% of positive results correspond to true disease. This counterintuitive result is a direct consequence of the low base rate, illustrating why prior probabilities matter enormously.

3.2 Bayesian Spam Filtering

A spam filter estimates $P(\text{spam} \mid \text{words in email})$. Using a naive Bayes classifier, it assumes word occurrences are conditionally independent given the class, then applies:

$$P(\text{spam} \mid w_1, \dots, w_n) \propto P(\text{spam}) \prod_{i=1}^n P(w_i \mid \text{spam})$$

Training data provide the likelihoods $P(w_i \mid \text{spam})$ and $P(w_i \mid \text{ham})$, and the prior $P(\text{spam})$ is estimated from the proportion of spam in the training set.

4. The Bayesian vs. Frequentist Debate

Frequentists interpret probability as the long-run frequency of events; they do not assign probabilities to hypotheses. Bayesians treat probability as a degree of belief, permitting statements like “there is a 73% chance this hypothesis is true.” The two schools differ in how they handle prior information:

- **Frequentists** rely solely on the likelihood function and avoid priors.
- **Bayesians** explicitly encode prior knowledge, which is then updated by data.

In practice, both approaches are valuable; the Bayesian framework is especially useful when prior knowledge is available or when sequential updating of beliefs is required.

5. Conclusion

Bayes’ theorem is a deceptively simple equation with profound implications. It provides a principled mechanism for learning from evidence: start with a prior belief, observe data, compute the likelihood, and obtain a posterior belief. Applications span medicine, finance, natural language processing, robotics, and fundamental science. As datasets grow and computational power increases, Bayesian methods continue to gain prominence as a rigorous framework for inference under uncertainty.